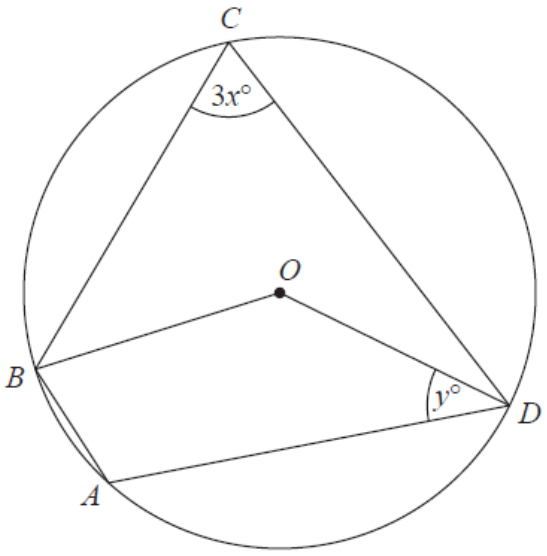
17



A, B, C and *D* are points on a circle with centre *O*.

size of angle *ABO* : size of angle *ODA* = 2 : 1

Find an expression for *y* in terms of *x*.
You must show all your working.

Angle BOD = 2 × 3x = 6x.

1 mark
Same arc BD.

Angle at centre is twice angle at circumference.

Angle BAD = 180° − 3x.

1 mark
Use cyclic ABCD.

Opposite angles in a cyclic quadrilateral total 180°.

ABO = 2y; in quadrilateral ABOD:

2y + y + 6x + (180 − 3x) = 360

1 mark
All four interior angles.

3y + 3x = 180

y = 60 − x

1 mark
y = 60 − x.

.....

(Total for Question 17 is 4 marks)